Azure Data Factory: Copy Files Matching a Specific Name Pattern

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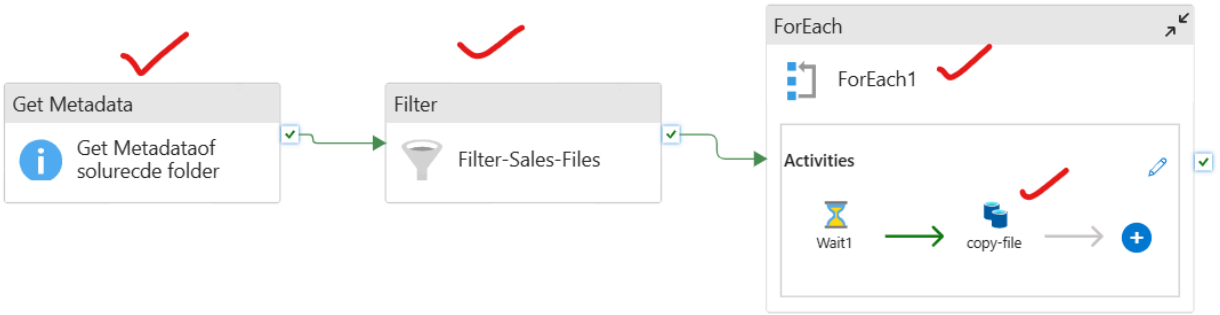

Scenario Overview

Scenario: Copy Only Daily Sales Files Matching a Date Pattern

A vendor uploads multiple types of files daily into a single **Azure Blob Storage** `landing` container:

```text
landing-or-source/
  ├── sales_20260423.csv        ← ✅ COPY THIS
  ├── sales_20260422.csv        ← ✅ COPY THIS
  ├── returns_20260423.csv      ← ❌ SKIP
  ├── inventory_20260423.json   ← ❌ SKIP
  └── sales_summary.xlsx        ← ❌ SKIP
```

Goal: Copy **only** files whose names match the pattern `sales_*.csv` from the `landing` container to the `processed` container — ignoring all other files.

Why this matters:
In real pipelines, landing zones are shared and noisy. Pattern-based filtering ensures your pipeline picks up **exactly the right files** without manual intervention or risky broad copies.

🏗️ Architecture

```text
[Azure Blob Storage : landing-or-source/]
  ├── sales_20260423.csv    ──┐
  ├── sales_20260422.csv    ──┼──► Filter by pattern: sales_*.csv
  ├── returns_20260423.csv  ──┤                  │
  ├── inventory_*.json      ──┘            (only matches)
  └── sales_summary.xlsx                         │
                                                 ▼
                                   [Get Metadata - childItems]
                                                 │
                                                 ▼
                                     [Filter Activity]
                                   @startsWith(item().name,'sales_')
                                   @endsWith(item().name,'.csv')
                                                 │
                                                 ▼
                                     [ForEach Activity]
                                                 │
                                                 ▼
                                     [Copy Activity per file]
                                                 │
                                                 ▼
                                [Azure Blob Storage : processed/]
                                  ├── sales_20260423.csv
                                  └── sales_20260422.csv
```

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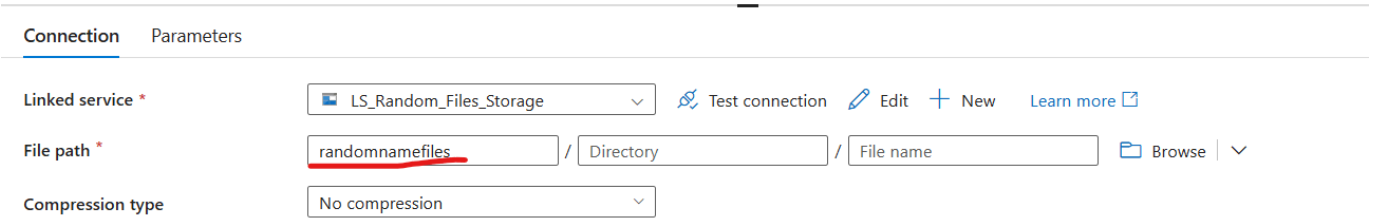
rerequisites

- **Azure Data Factory** instance (v2)
- **Azure Blob Storage** with two containers:
  - `landing` — shared drop zone with mixed file types
  - `processed` — clean destination for matched files only
- Linked Service: `LS_AzureBlobStorage`

Step 1: Create Datasets

1a. Landing-or-source Container Dataset (for Get Metadata — Folder Level)

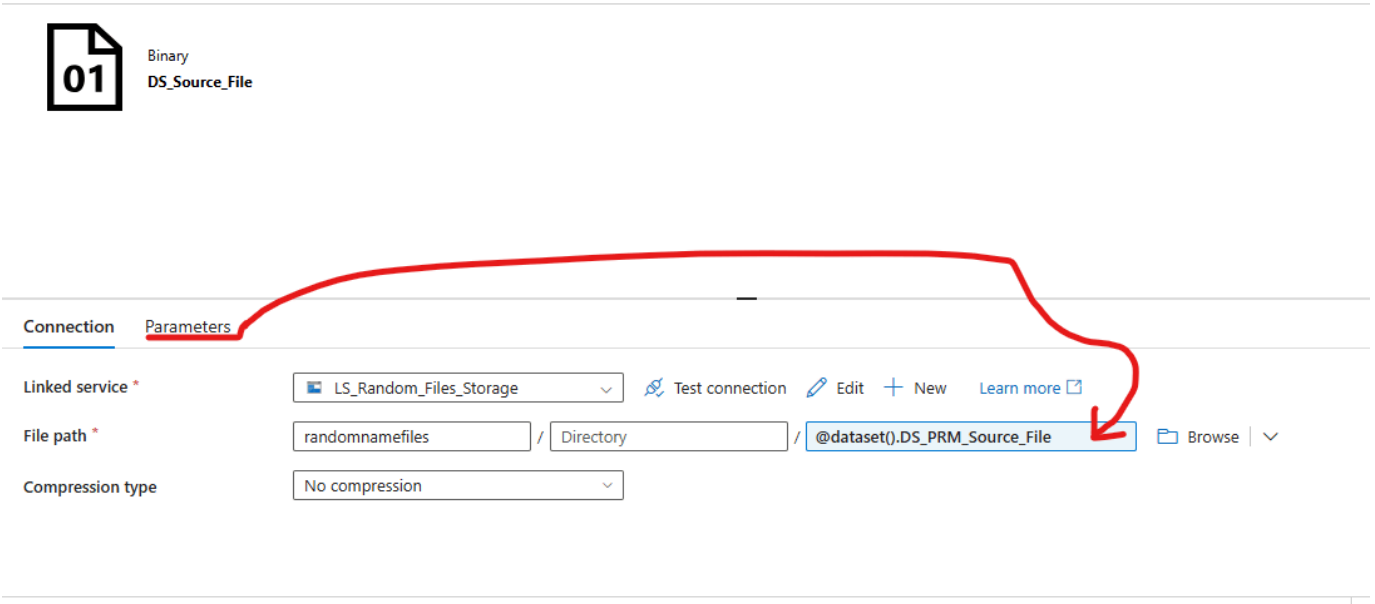
1. Go to **Author** → **Datasets** → **+ New Dataset**
2. Select **Azure Blob Storage** → **Binary**
3. Name it `DS_Landing_Folder`
4. Linked Service: `LS_AzureBlobStorage`
5. **Connection tab:**
    - Container: `landing`
    - Leave **Directory** and **File** blank (we're pointing at the folder)
6. **Publish All**

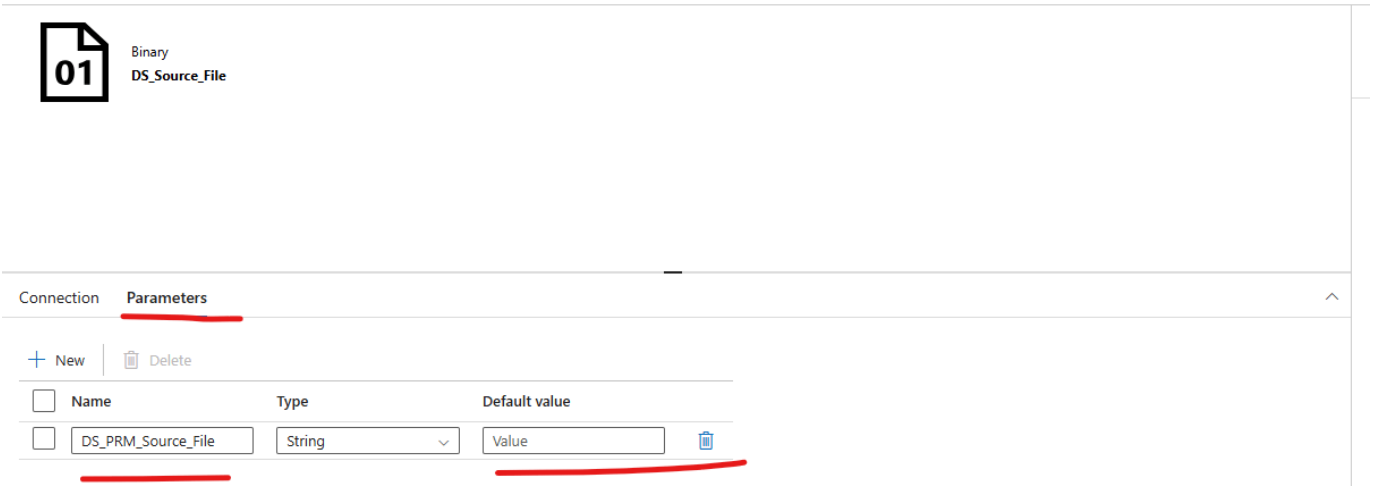


1b. Landing-or-Source File Dataset (Parameterized — for Copy Activity Source)

1. **+ New Dataset** → Azure Blob Storage → **Binary**
2. Name it `DS_Landing_File`
3. Linked Service: `LS_AzureBlobStorage`
4. **Parameters tab** → Add:
    - `fileName` | Type: `String`

5. **Connection tab:**
    - Container: `landing`
    - File name → Add dynamic content: `@dataset().fileName`
6. **Publish All**



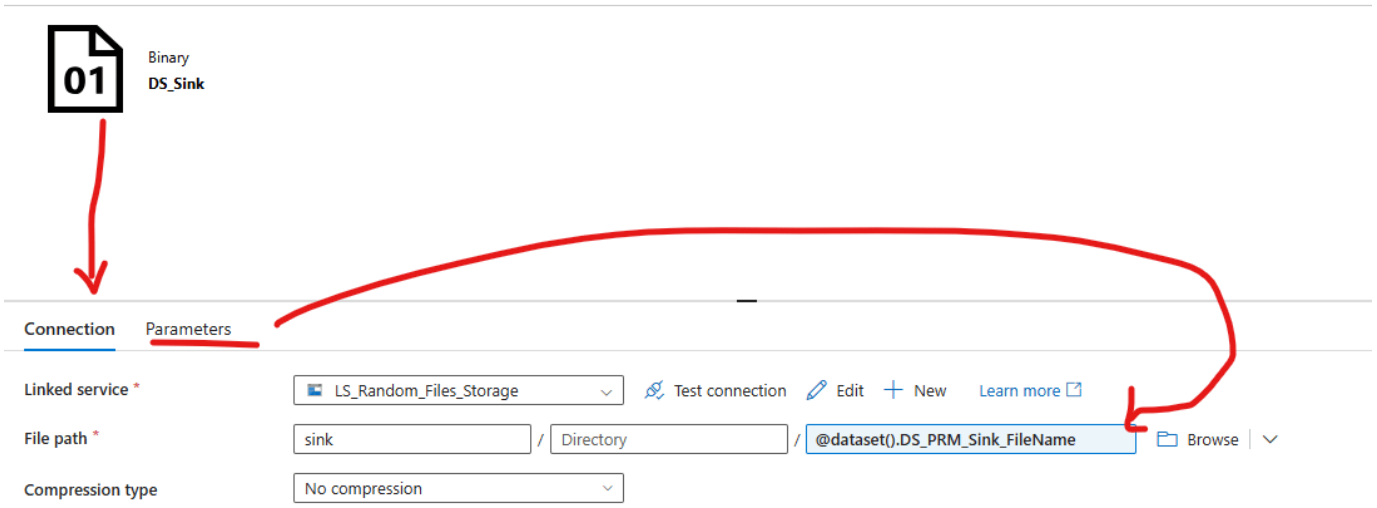


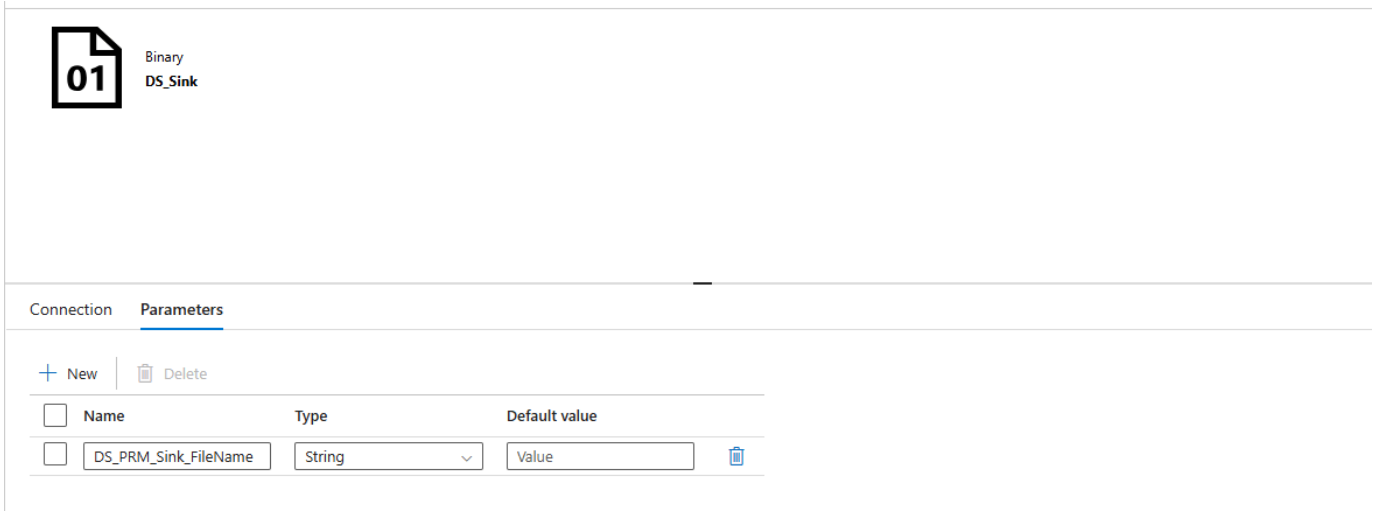
1c. Processed-or-Destination Container Dataset (Parameterized — for Copy Activity Sink)

1. **+ New Dataset** → Azure Blob Storage → **Binary**
2. Name it `DS_Processed_File`
3. Linked Service: `LS_AzureBlobStorage`
4. **Parameters tab** → Add:
    - `fileName` | Type: `String`

5. **Connection tab:**
    - Container: `processed`
    - File name → Add dynamic content: `@dataset().fileName`
6. **Publish All**

> **Why Binary datasets?**
> Binary datasets copy files as-is without parsing content — perfect for file movement regardless of format (CSV, JSON, Parquet, etc.).



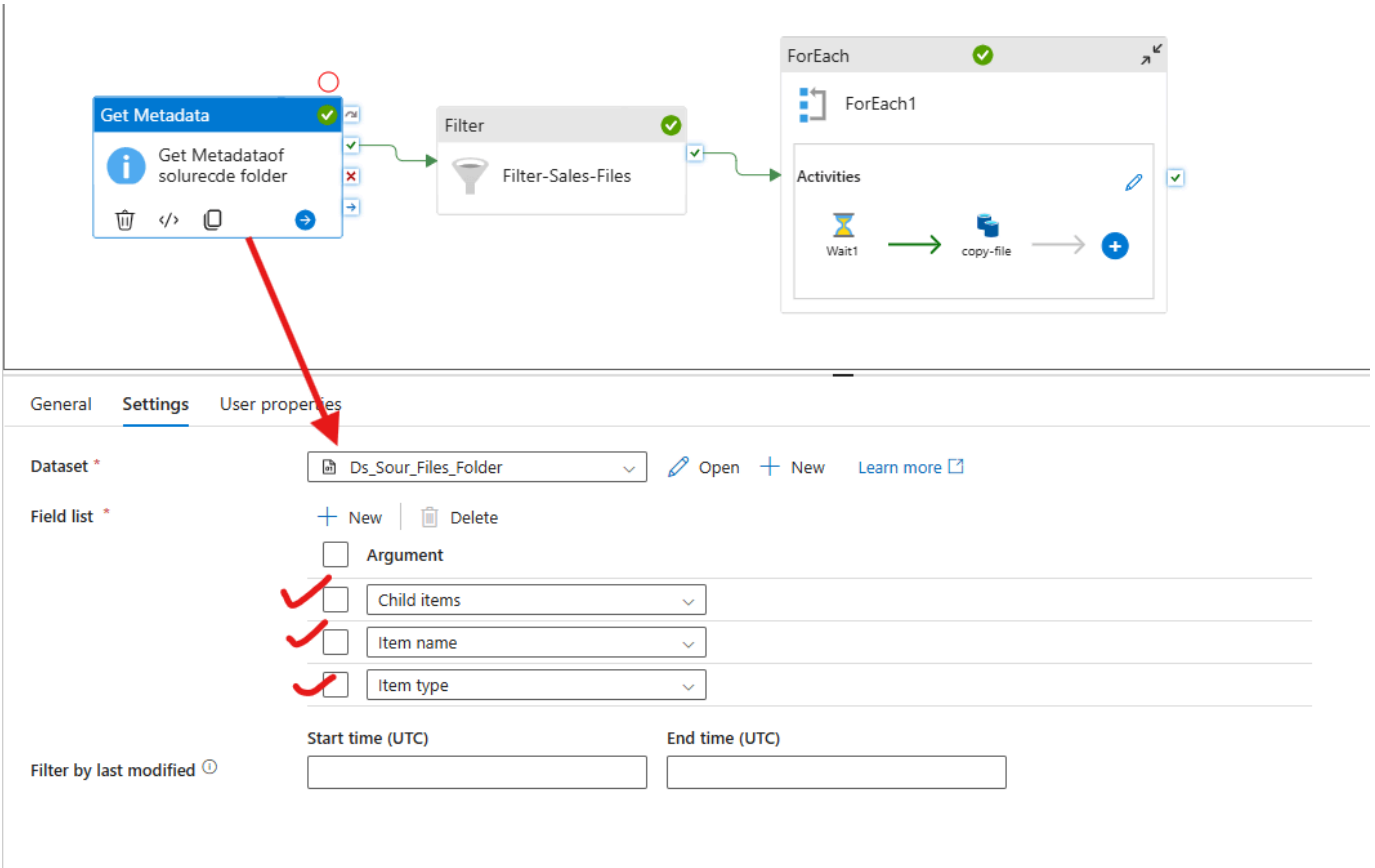


Step 2: Add Get Metadata Activity (List All Files in Folder)

1. Go to **Author** → **Pipelines** → **+ New Pipeline**
2. Name it `PL_CopyPatternMatchedFiles`

3. Drag **Get Metadata** onto the canvas
4. Name it `GMT_ListLandingFiles`
5. Click the activity → **Settings** tab:
    - **Dataset:** `DS_Landing_Folder`
    - **Field list** → Click **+ New** → Select: `childItems`

> `childItems` returns an **array** of all items (files and folders) in the container. Each item in the array has two properties:
>
> - `name` — the file name string (e.g., `sales_20260423.csv`)
> - `type` — either `File` or `Folder`



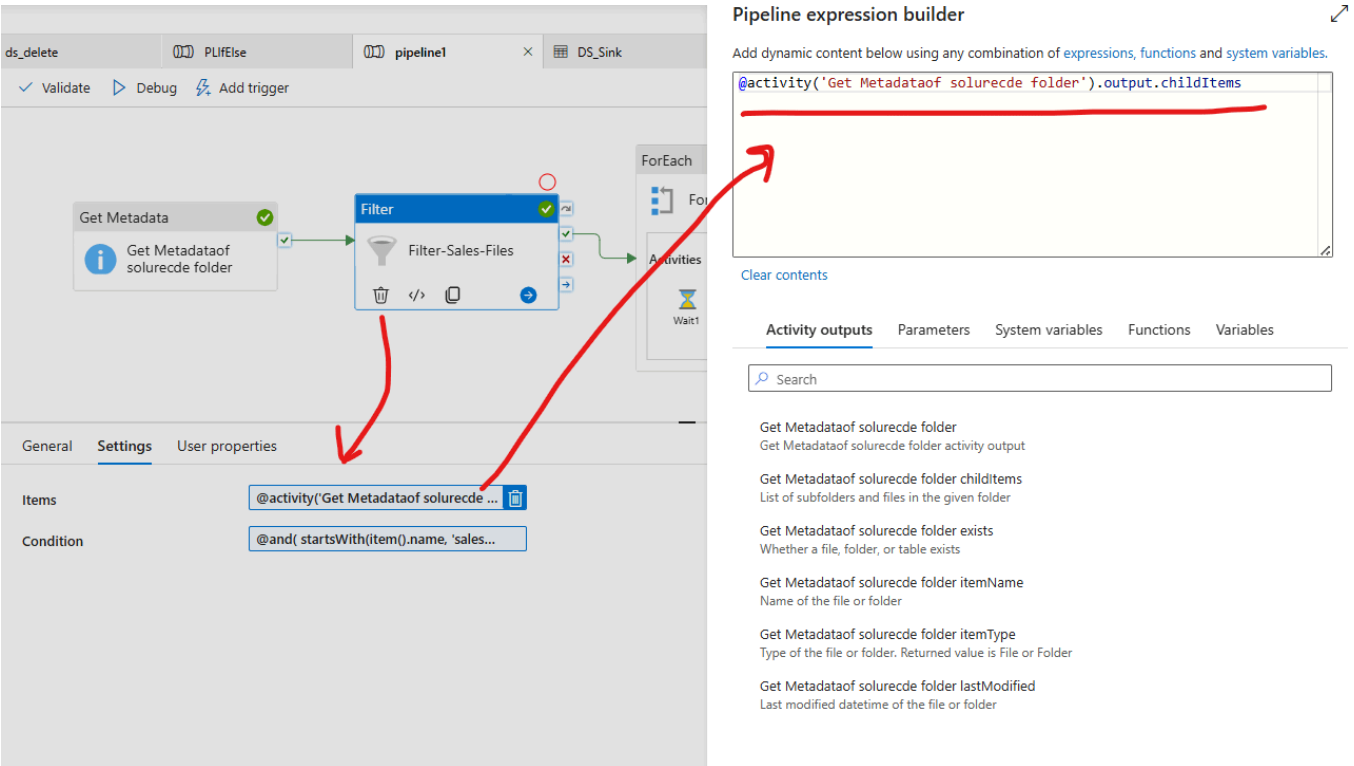
Step 3: Add Filter Activity (Apply the Name Pattern)

1. Drag a **Filter** activity onto the canvas
2. Name it `FLT_MatchSalesCSV`
3. Draw a **Success** arrow: `GMT_ListLandingFiles` → `FLT_MatchSalesCSV`
4. Click the activity → **Settings** tab:

Items field:

Click **Add dynamic content** and enter:

```text
@activity('GMT_ListLandingFiles').output.childItems
```

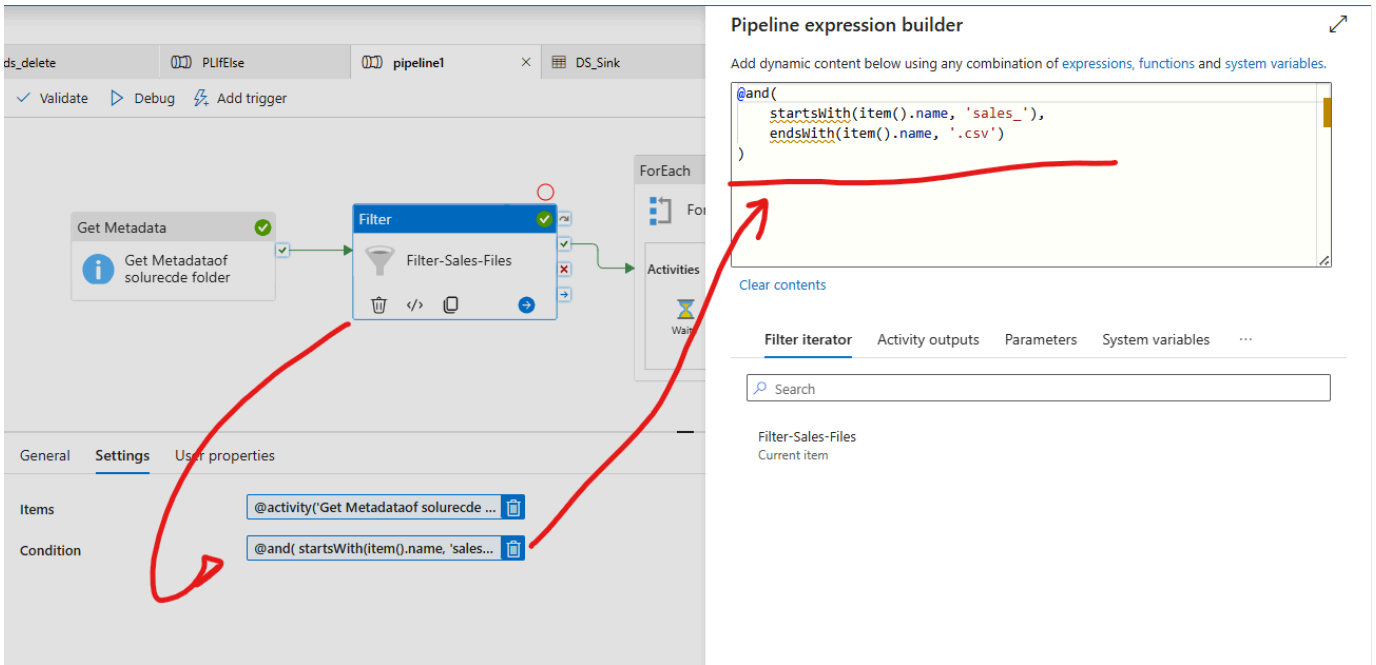


Condition field:

Click **Add dynamic content** and enter:

```text
@and(
    startsWith(item().name, 'sales_'),
    endsWith(item().name, '.csv')
)
```

> **What this does:**
>
> - `startsWith(item().name, 'sales_')` → keeps only files beginning with `sales_`
> - `endsWith(item().name, '.csv')` → keeps only files ending with `.csv`
> - `@and(...)` → BOTH conditions must be true — eliminates `sales_summary.xlsx`

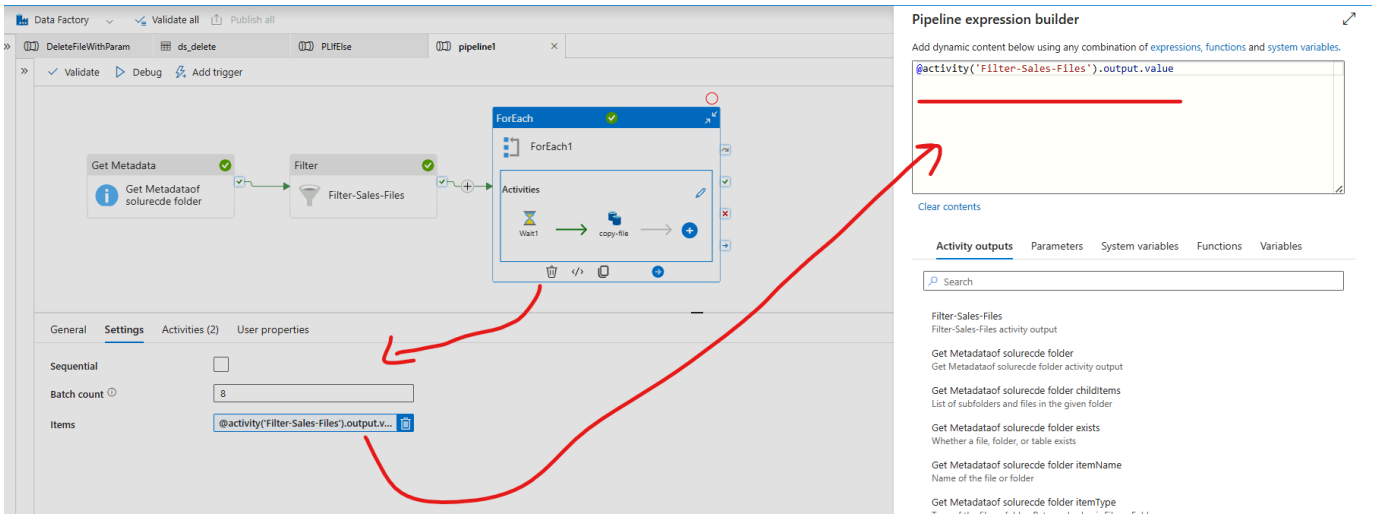


🔄 Step 4: Add ForEach Activity (Iterate Matched Files)

1. Drag **ForEach** onto the canvas
2. Name it `FE_IterateMatchedFiles`
3. Draw a **Success** arrow: `FLT_MatchSalesCSV` → `FE_IterateMatchedFiles`
4. Click the activity → **Settings** tab:
    - **Sequential:** OFF (run in parallel)
    - **Batch count:** `4`
    - **Items** → Add dynamic content:

```text
@activity('FLT_MatchSalesCSV').output.Value
```

> Note: The Filter activity output property is `.output.Value` (capital **V**), unlike Lookup which uses `.output.value` (lowercase v).

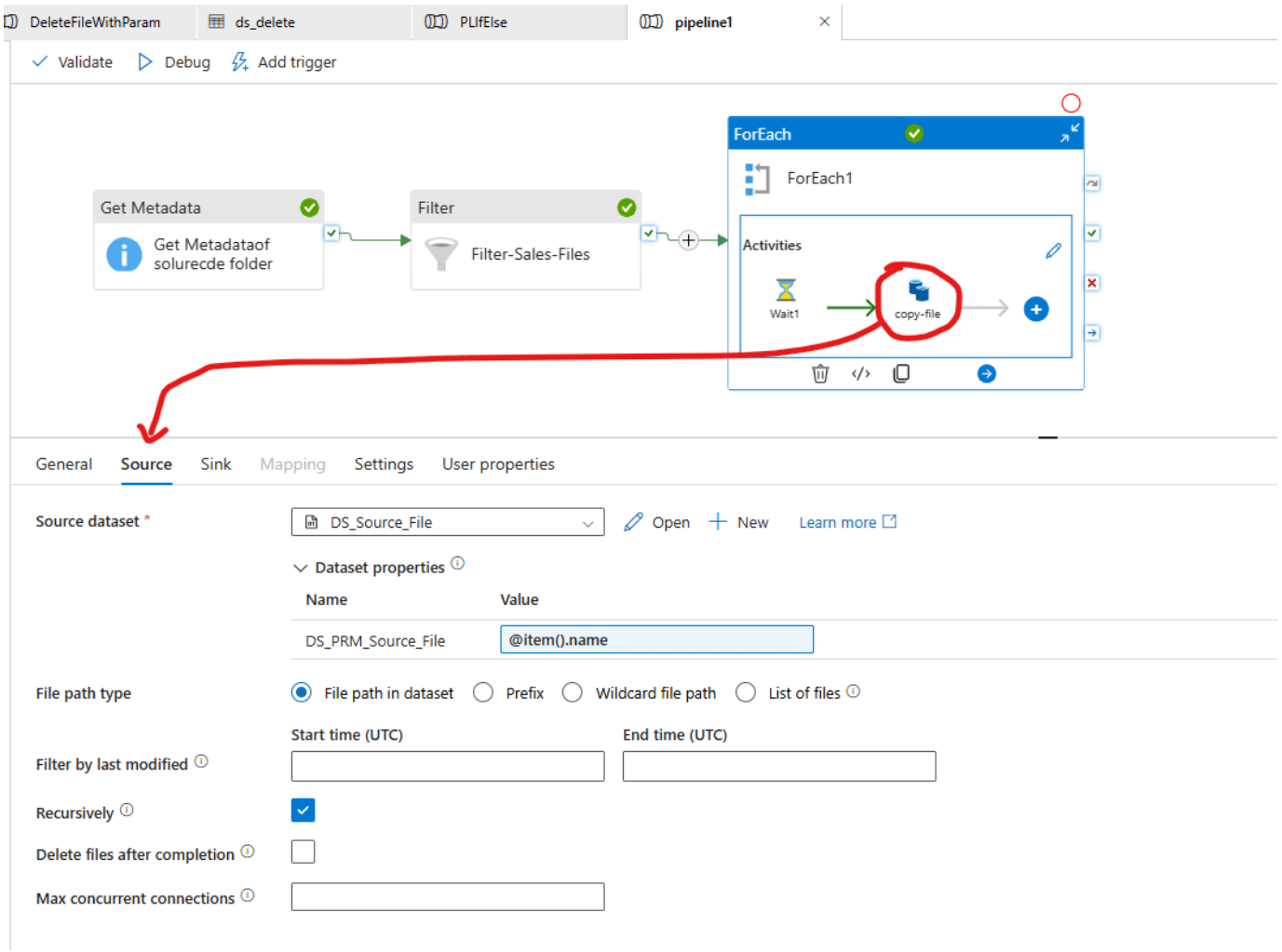


📋 Step 5: Add Copy Activity Inside ForEach

1. Click the **pencil icon (✏️)** on `FE_IterateMatchedFiles` to open the inner canvas
2. Drag **Copy Data** onto the inner canvas
3. Name it `CPY_CopyMatchedFile`

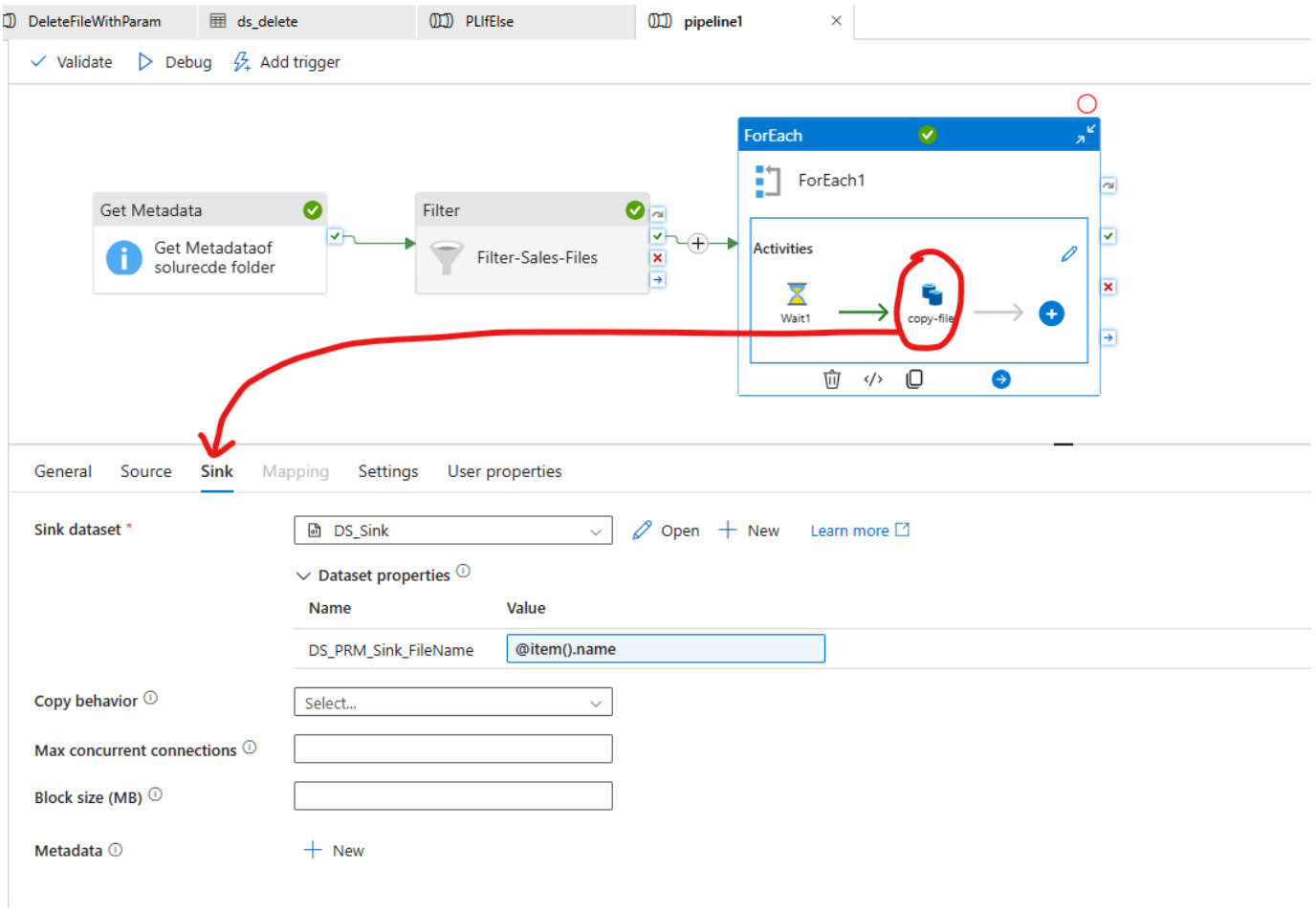
Source Tab:

- **Dataset:** `DS_Landing_File`
- **Dataset properties:**
  - `fileName` → Add dynamic content: `@item().name`



Sink Tab:

- **Dataset:** `DS_Processed_File`
- **Dataset properties:**
  - `fileName` → Add dynamic content: `@item().name`



Settings Tab (Optional):

- Enable **Delete source files after copy** → converts copy into a **move** operation, cleaning up the landing zone

4. Click ← **back arrow** to return to the main pipeline canvas

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Step 6: Debug and Validate

Setup test files in `landing` container:

File Name	Should Be Copied?
`sales_20260423.csv`	✅ Yes — matches `sales_*.csv`
`sales_20260422.csv`	✅ Yes — matches `sales_*.csv`
`returns_20260423.csv`	❌ No — wrong prefix
`inventory_20260423.json`	❌ No — wrong prefix and extension
`sales_summary.xlsx`	❌ No — wrong extension

File Name	Should Be Copied?
`SALES_20260421.csv`	❌ No — `startsWith` is case-sensitive

Debug Steps:

1. Click **Debug** in the pipeline toolbar
2. Monitor **Output** panel:
   - `GMT_ListLandingFiles` → output shows 5 `childItems`
   - `FLT_MatchSalesCSV` → output shows 2 filtered items
   - `FE_IterateMatchedFiles` → runs 2 iterations
   - `CPY_CopyMatchedFile` → copies 2 files
3. Verify `processed` container now contains:
   - `sales_20260423.csv` ✅
   - `sales_20260422.csv` ✅

Full Pipeline View Summary

```text
Pipeline: PL_CopyPatternMatchedFiles

[GMT_ListLandingFiles]
  Get Metadata → childItems (all files in landing/)
        │
        ▼ (on success)
[FLT_MatchSalesCSV]
  Filter → @and(startsWith(name,'sales_'), endsWith(name,'.csv'))
        │
        ▼ (on success)
[FE_IterateMatchedFiles]
  ForEach → @activity('FLT_MatchSalesCSV').output.Value
        │
        └─► [CPY_CopyMatchedFile]
              Source: DS_Landing_File  → @item().name
              Sink:   DS_Processed_File → @item().name
```

Pattern Variation Examples

Different business needs call for different pattern expressions. Here are ready-to-use Filter conditions:

Match files from today only

```text
@and(
    startsWith(item().name, 'sales_'),
    endsWith(item().name, '.csv'),
    endsWith(item().name, concat(formatDateTime(utcNow(), 'yyyyMMdd'), '.csv'))
)
```

Match any CSV file (any prefix)

```text
@endsWith(item().name, '.csv')
```

Match files containing a keyword anywhere in the name

```text
@and(
    contains(item().name, 'sales'),
    endsWith(item().name, '.csv')
)
```

Match files by multiple valid extensions

```text
@and(
    startsWith(item().name, 'sales_'),
    or(
        endsWith(item().name, '.csv'),
        endsWith(item().name, '.parquet')
    )
)
```

Match files only (exclude subfolders)

```text
@and(
    equals(item().type, 'File'),
    startsWith(item().name, 'sales_'),
    endsWith(item().name, '.csv')
)
```

Match files modified today using Get Metadata inside ForEach

> Use a nested Get Metadata (inside ForEach) on each file with `lastModified` field,
> then wrap Copy inside another If Condition to check the date.

Key Expressions Reference

Expression	Purpose
`@activity('GMT_ListLandingFiles').output.childItems`	Array of all files/folders in the container
`@activity('FLT_MatchSalesCSV').output.Value`	Array of items that passed the filter (capital V)
`@item().name`	File name of the current ForEach iteration
`@item().type`	`"File"` or `"Folder"` for current iteration item
`startsWith(item().name, 'sales_')`	Checks if name begins with `sales_`
`endsWith(item().name, '.csv')`	Checks if name ends with `.csv`

Expression	Purpose
`contains(item().name, 'sales')`	Checks if name contains `sales` anywhere
`@and(expr1, expr2)`	Both conditions must be true
`@or(expr1, expr2)`	Either condition must be true
`formatDateTime(utcNow(), 'yyyyMMdd')`	Today's date as `20260423`

Common Pitfalls to Avoid

- ❌ **Case sensitivity** — `startsWith('SALES_20260423.csv', 'sales_')` returns `false`. Use `toLower(item().name)` to normalize: `startsWith(toLower(item().name), 'sales_')`
- ❌ **Using `.output.value` (lowercase)** for Filter output — Filter uses `.output.Value` with a capital **V**. Using lowercase causes a null reference error.
- ❌ **Pointing Get Metadata at a file instead of a folder** — The `childItems` field only works when the dataset points to a container or directory, not a specific file.
- ❌ **Not filtering by `type = 'File'`** — If subfolders exist in `landing/`, they appear in `childItems` too. Add `equals(item().type, 'File')` to your Filter condition to avoid errors.
- ❌ **Skipping the Filter activity** — Running ForEach directly on unfiltered `childItems` will attempt to copy every file, including non-target files, causing failures.

Extension Ideas

Enhancement	How to Implement
Archive after copy	Enable **Delete source files** in Copy Activity sink settings

Enhancement	How to Implement
Log copied file names	Add a stored procedure or Append Variable activity after Copy to log each file
Alert if no files matched	Add an **If Condition** after Filter: `@equals(length(activity('FLT_MatchSalesCSV').output.Value), 0)`
Process files from a specific date range	Add `contains(item().name, '202604')` to filter by month prefix
Handle nested subfolders	Use a recursive approach with multiple Get Metadata calls or enable recursive copy in the Copy Activity settings

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